

SOLUTION DOCUMENT

Custom Web-Based Learning Management System (LMS)

Prepared for:

Client: DSA Forge

Prepared by:

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Confidential - Prepared exclusively for the addressed recipient

1. Document Control

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2. Executive Summary

This document presents the proposed solution for a custom, web-based Learning Management System (LMS) designed for institutes and universities. The platform will bring course delivery, examinations, and hands-on coding assessments together under a single portal, with dedicated experiences for Admin, Faculty/Trainer, and Student users.

The solution is organized around three core modules: Courses, Exams, and a Coding Portal - all operating within one unified LMS, governed by role-based access so each user sees only what is relevant to them.

3. Project Objectives

- Provide a single, centralized platform for course delivery, examinations, and coding practice/assessment.
- Give administrators complete visibility and control over users, courses, and institute-wide activity.
- Enable faculty to create courses, assign tasks, and schedule exams with minimal manual effort.
- Give students a simple, self-service experience to enroll, learn, attempt exams, and practice coding.
- Build a platform that is scalable and can grow with additional modules in future phases.

4. Scope of Work

Based on the requirements shared, the initial phase of the platform covers the following three core modules, to be delivered within a 1-month timeline:

- **Courses (LMS):** Course creation, content delivery, and enrollment management.
- **Exam Module (LMS):** Exam scheduling, question banks, and automated evaluation.

- **Coding Portal (LMS):** Online code editor with automated, test-case based evaluation.

All three modules will be accessible through role-based dashboards for Admin, Faculty/Trainer, and Student users, as detailed in the following sections.

5. User Roles & Permissions

The platform supports three primary user roles, each with a distinct dashboard and permission set:

Role	Core Responsibilities
Admin	- Owns and controls the overall portal configuration and settings - Creates and manages Faculty/Trainer and Student accounts - Approves and oversees courses, exams, and coding modules - Manages roles, permissions, and access control - Views institute-wide reports, analytics, and usage dashboards - Sends portal-wide announcements and notifications
Faculty / Trainer	- Creates and manages courses, modules, and learning content - Assigns tasks and assignments to enrolled students - Schedules exams and builds question banks - Creates coding problems/challenges for the Coding Portal - Evaluates and grades student submissions - Tracks individual and batch-wise student progress
Student	- Registers and creates a profile on the portal - Enrolls in courses assigned or made available to them - Accesses course content and submits assignments - Participates in scheduled exams within the given window - Attempts coding problems on the Coding Portal - Views results, scorecards, and performance history

5.1 Typical User Journeys

Admin

- Logs in → sets up Faculty and Student accounts → configures courses and institute-wide settings → monitors activity via dashboards and reports.

Faculty / Trainer

- Logs in → creates a course and uploads content → assigns tasks to enrolled students → schedules an exam or coding challenge → reviews submissions and grades.

Student

- Registers on the portal → enrolls in an assigned course → studies content and submits assignments → attempts scheduled exams and coding challenges → views results.

6. Module 1 — Courses (LMS)

Enables Faculty/Trainers to build structured courses and Students to consume them at their own pace.

Key Features

- Course creation with modules, chapters, and lessons
- Support for multiple content formats — video, PDF, PPT, and documents
- Course categorization by department, subject, or level
- Manual or auto-enrollment of students into courses
- Assignment creation, submission, and tracking
- Student progress tracking (per course and per lesson)
- Optional certificate generation on course completion

7. Module 2 — Exam Module (LMS)

Allows Faculty/Trainers to schedule and manage exams, with automated evaluation wherever possible.

Key Features

- Exam creation and scheduling with defined date, time window, and duration
- Question bank management - MCQ, true/false, and subjective question types
- Randomized questions and answer options to reduce malpractice
- Auto-submit when the time limit is reached
- Automatic grading for objective questions; manual grading workflow for subjective ones
- Result and scorecard generation, visible to both Faculty and Students
- Basic exam-integrity checks (e.g., tab-switch / full-screen exit detection) — scope to be finalized with client

8. Module 3 — Coding Portal (LMS)

Gives students a hands-on environment to practice and be evaluated on programming problems set by Faculty.

Key Features

- Faculty-created coding problems with description, constraints, and sample test cases
- Browser-based code editor supporting multiple programming languages
- Real-time code compilation and execution in a sandboxed environment
- Automated evaluation against hidden/visible test cases
- Submission history for each student, per problem
- Optional leaderboard to encourage participation

9. Non-Functional Requirements

- **Security:** Role-based access control, encrypted credentials, and secure session handling.
- **Scalability:** Architecture designed to support a growing number of concurrent users, courses, and exams.
- **Responsiveness:** Fully usable across desktop, tablet, and mobile browsers.
- **Availability:** Cloud-hosted with regular automated backups.
- **Performance:** Optimized for smooth performance during peak load, such as concurrent exam attempts.
- **Audit & Logs:** Activity logs for key actions (exam attempts, submissions, admin changes).

10. Proposed Technology Stack

The stack below is an initial recommendation and can be adjusted based on the client's existing infrastructure or preferences.

Layer	Proposed Technology
Frontend (Web)	React.js with a responsive UI framework (mobile & tablet friendly)
Backend / API	Node.js (Express) — alternatively Django / Laravel based on team preference
Database	PostgreSQL / MySQL for core data; Redis for caching & session management
Code Execution Engine	Sandboxed, Docker-based judge service (e.g., Judge0-style) for the Coding Portal
Authentication	JWT-based session tokens with role-based access control (RBAC)
Hosting / Infra	Cloud hosting (AWS / Azure / GCP) with automated backups
Notifications	Email (SMTP/SES) and in-portal notifications; SMS optional

11. High-Level Architecture Overview

The platform will follow a modular, layered architecture so that the Courses, Exam, and Coding Portal modules can be developed, scaled, and maintained independently while sharing a common user, authentication, and notification layer:

- **Presentation Layer:** responsive web application used by Admin, Faculty, and Students
- **Application/API Layer:** business logic for courses, exams, coding evaluation, and user management
- **Data Layer:** relational database for core records, with caching for performance
- **Code Execution Layer:** isolated sandbox service for compiling and running student code securely

- Notification Layer: email and in-portal alerts for enrollments, exam schedules, and results

12. Project Timeline

Based on the discussed target of a 1-month delivery, the work is proposed to be executed across the following phases. This plan is indicative and will be finalized after detailed requirement sign-off.

Week	Focus Area	Key Activities
Week 1	Foundation & Admin Panel	Requirement sign-off & UI/UX wireframes Admin panel: user & role management setup Authentication and access-control framework
Week 2	Courses Module	Course creation, content upload, and categorization Student enrollment workflow Faculty assignment & task-management features
Week 3	Exam Module & Coding Portal (Start)	Exam creation, scheduling, and question bank Auto-evaluation for objective questions Coding Portal: online editor and problem setup begins
Week 4	Coding Portal, Integration & Go-Live	Coding Portal: test-case based auto-evaluation End-to-end integration across all three modules UAT with client, bug fixes, and deployment

13. Assumptions & Dependencies

- Client will provide timely feedback and content (sample courses, question sets, branding assets) during development.
- Final list of supported programming languages for the Coding Portal will be confirmed during requirement finalization.
- Hosting/cloud account access and any required third-party service credentials (email/SMS gateway) will be provided by the client or procured jointly.
- Advanced proctoring (AI-based camera monitoring, biometric checks) is not included in this phase unless explicitly scoped.
- The 1-month timeline assumes scope remains within the three modules described in this document.

14. Out of Scope (Phase 1)

- Web platforms will be responsive and mobile-friendly instead.
- Advanced AI-based proctoring and plagiarism detection.
- Payment gateway integration for paid courses, unless required and scoped separately.
- Integration with third-party university ERP/SIS systems, unless specified.

15. Deliverables

- Fully functional web-based LMS platform with Admin, Faculty/Trainer, and Student portals
- Courses, Exam, and Coding Portal modules as described in Sections 6–8
- Basic user documentation / walkthrough for each role
- Deployment on agreed cloud environment
- Post-launch support window (duration to be mutually agreed)

